

Experiment: Inverting and Non-Inverting Amplifiers

AIM

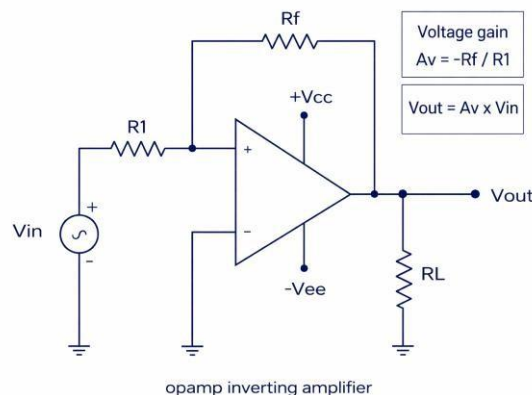
To design and realize inverting and non-inverting amplifiers using the IC 741 operational amplifier and verify their voltage gain and phase characteristics.

APPARATUS REQUIRED

- Breadboard
- IC 741 Operational Amplifier
- Dual Power Supply ($\pm 12\text{ V}$)
- Resistors ($1\text{ k}\Omega$, $10\text{ k}\Omega$, $100\text{ k}\Omega$)
- Function Generator
- CRO/DSO
- Connecting leads

THEORY

An operational amplifier (Op-Amp) is a high gain differential amplifier with very high input impedance and very low output impedance. By using negative feedback, stable and accurate voltage gain can be obtained. The two most common configurations are the inverting and non-inverting amplifiers.



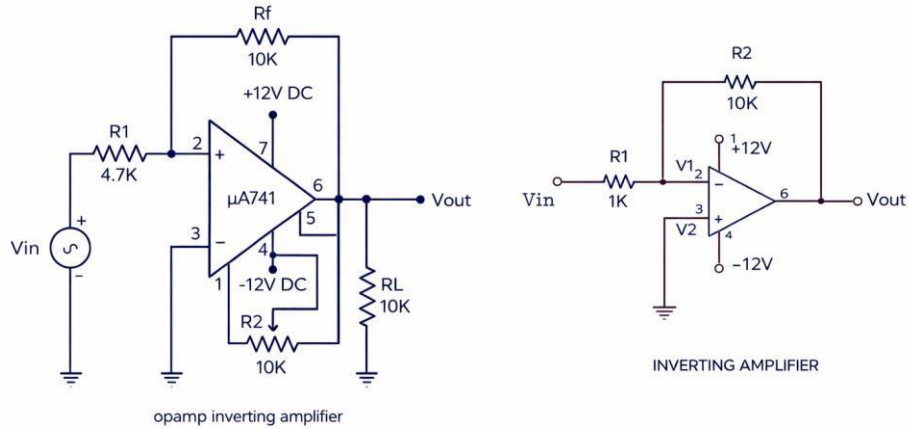
INVERTING AMPLIFIER

In this configuration the input signal is applied to the inverting terminal through an input resistor R_1 and the non-inverting terminal is grounded. A feedback resistor R_f connects the output to the inverting input. The output is amplified and 180° out of phase with the input.

Voltage Gain: $A_v = - R_f / R_1$

Characteristics

- Output is phase inverted
- Gain depends on R_f and R_1
- Virtual ground at inverting terminal



Design Values

For Gain = 1 : $R_1 = 10\text{ k}\Omega$, $R_f = 10\text{ k}\Omega$

For Gain = 10 : $R_1 = 10\text{ k}\Omega$, $R_f = 100\text{ k}\Omega$

Observation Table (Inverting)

Vin (V)	Vout (V)
0	
0.1	
0.3	
0.5	
0.7	
0.9	

NON-INVERTING AMPLIFIER

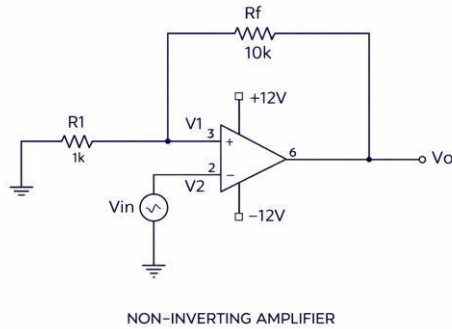
Here the input is applied to the non-inverting terminal and feedback is provided to the inverting terminal. The output is in phase with the input and has very high input impedance.

Voltage Gain: $A_v = 1 + (R_f / R_1)$

Characteristics

- No phase shift
- High input impedance

- Gain greater than or equal to 1



Design Values

For Gain = 2 : $R_1 = 10 \text{ k}\Omega$, $R_f = 10 \text{ k}\Omega$

For Gain = 10 : $R_1 = 10 \text{ k}\Omega$, $R_f = 90 \text{ k}\Omega$

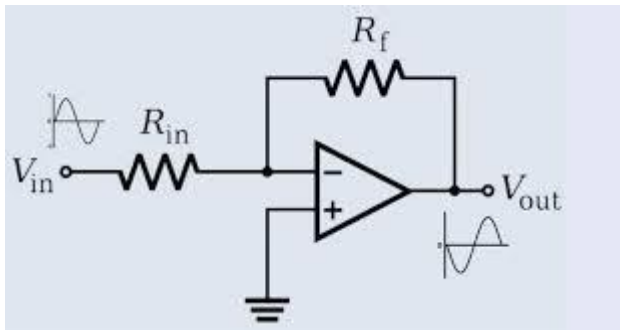
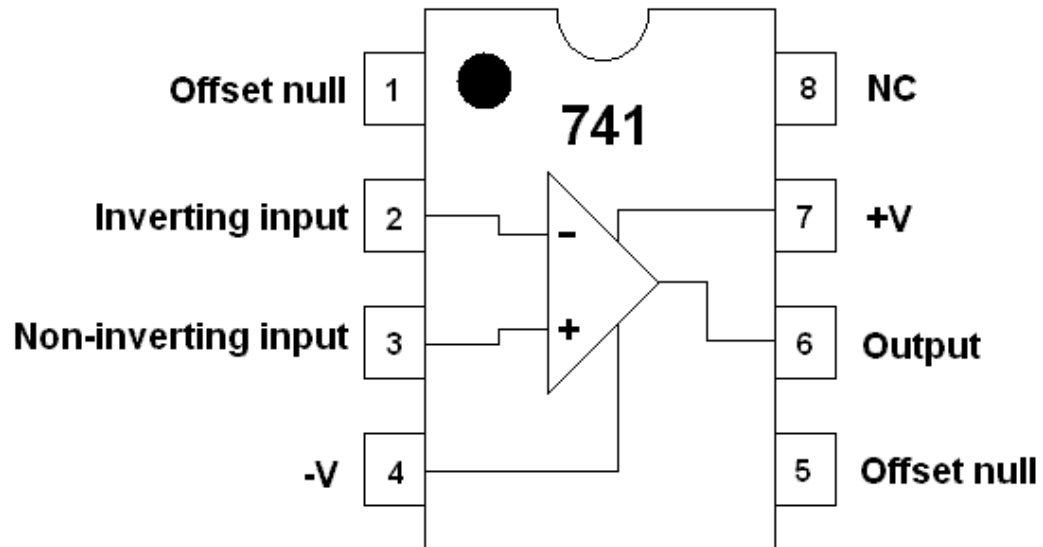
Observation Table (Non-Inverting)

Vin (V)	Vout (V)
0	
0.1	
0.3	
0.5	
0.7	
0.9	

PROCEDURE

1. Connect the circuit for the inverting amplifier on the breadboard.
2. Apply $\pm 12 \text{ V}$ supply to the IC 741.
3. Feed input signal from the function generator.
4. Measure input and output using CRO/DSO.
5. Repeat the procedure for the non-inverting configuration.
6. Record readings and calculate gain.

CORRECT INVERTING AMPLIFIER CONNECTION (IC 741)



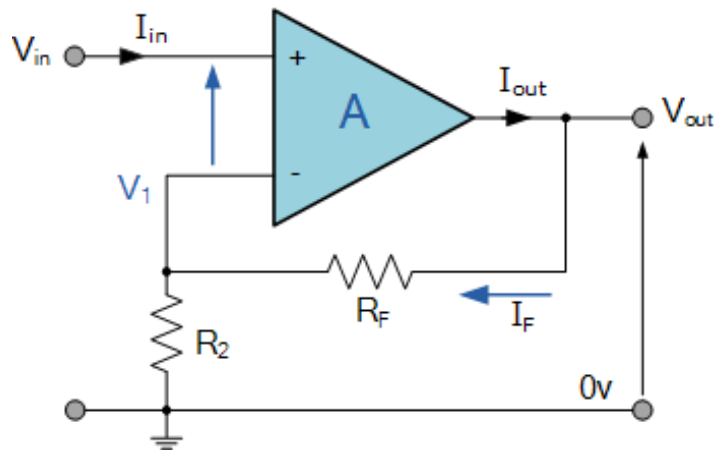
Correct Pin Connections:

- **Pin 2** → Inverting input (-)
- **Pin 3** → Non-inverting input (+) → Ground
- **Pin 6** → Output
- **Pin 7** → +12V
- **Pin 4** → -12V

Correct Circuit:

- $V_{in} \rightarrow R_{in} \rightarrow \text{Pin 2}$
- $R_f \rightarrow \text{Between Pin 6 and Pin 2}$
- $\text{Pin 3} \rightarrow \text{Ground}$

CORRECT NON-INVERTING AMPLIFIER



- Connect **V_{in}** to **Pin 3** (Non-inverting input).
- Connect **R_f** between **Pin 6** and **Pin 2**.
- Connect **R_1** between **Pin 2** and **Ground**.
- Take **Output** from **Pin 6**.
- Connect **Pin 7** to **+12V** supply.
- Connect **Pin 4** to **-12V** supply.

VIVA VOCE QUESTIONS

7. Define an operational amplifier.
8. What is virtual ground?
9. Why is the output inverted in inverting amplifier?
10. State advantages of non-inverting amplifier.
11. What is the operating voltage range of IC 741?